

MD. SOHEL RANA

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📍 Dhaka, Bangladesh

PROFESSIONAL SUMMARY

Final-year CSE student with strong problem-solving skills and hands-on experience building backend systems, cloud-assisted applications, and ML-powered products.

EDUCATION

BSc in Computer Science & Engineering

Bangladesh University of Business and Technology - BUBT

Coursework: Data Structures, Algorithms, OOP, DBMS

CGPA: 3.52/4.00

Expected Graduation: December 2026

PROBLEM SOLVING ACTIVITIES

- Codeforces: [Mr_Sohel](#) — Rating: 1422 (Specialist, Top 15%)
- CodeChef: [Mr_Sohel](#) — Rating: 1623 (3 Star Coder, Top 7%)
- LeetCode: [Mr_Sohel](#) — Rating: 1653 (Knight, Top 17.43%)

Solved over **1500+** algorithmic problems and participated in **140+** online contests.

[Problem Solving Statistics](#)

ACHIEVEMENTS & AWARDS

International & Regional

- **Meta Hacker Cup 2025, Round 2** [Standings](#)
Advanced to Round 2 among 13,779 global participants.
- **ICPC Asia Dhaka Regional 2025** [Standings](#)
Team: BUBT Mudbloods; onsite finalist among 1800+ nationwide teams; received **Honorable Mention**.
- **ICPC Asia Dhaka Regional 2024** [Standings](#)
Team: BUBT Mudbloods; ranked 157th nationally in the onsite finals.

University Contests

- **UAP Inter University Contest** [Standings](#)
Team: BUBT Mudbloods; achieved **2nd Runner-up** position among 80+ university teams.
- **BUBT IUPC (On Campus)** [Standings](#)
Team: BUBT Mudbloods; **2-Time Champion (2023, 2024)** among 100+ competing teams

SKILLS

Languages:

C/C++, C#, Python, JavaScript, SQL

Backend:

ASP.NET Core MVC, REST APIs, FastAPI

Frontend:

Angular, React.js

ML / Applied AI:

PyTorch, YOLOv8, CNNs, Transformers, ONNX Runtime

Core CS:

Data Structures, Algorithms, OOP, SOLID Principles, Design Patterns, DBMS

Tools:

Git, Docker, Linux,

PROJECTS

Real-Time Fruit Quality Detection

[Repo/Demo](#)

- **Problem & Contribution:** Built a real-time fruit quality detection system using ESP32-CAM, FastAPI, and YOLOv8 to enable low-cost automated inspection by designing an asynchronous edge-to-cloud pipeline that transmitted image payloads from edge devices to cloud-based inference services and delivered prediction results through a Telegram bot interface.
- **Result:** Improved inference latency by 80% by converting PyTorch weights to ONNX, reducing response time from 15s to 3s while achieving 98.97% mAP@50 across 8 fruit classes.
- **Stack:** Python, C++, FastAPI, YOLOv8 Nano, ONNX Runtime, Hugging Face, ESP32-CAM, Telegram API.

Facial Emotion Recognition for Autism [Repo/Demo](#)

- **Problem & Contribution:** Developed an end-to-end facial emotion recognition system focused on classifying subtle, atypical expressions for autism-assistive software. Designed and engineered a novel Enhanced SE-Swin Transformer (ESSwin-6) architecture to effectively process and model complex, multidimensional visual data.
- **Result:** Built a robust image preprocessing and dynamic augmentation pipeline to improve data quality and overall system reliability. Outperformed 14 standard baselines to achieve an 89% validation accuracy through rigorous model tuning, integrating Explainable AI to ensure transparent and debuggable predictions.
- **Stack:** Python, PyTorch, Vision Transformers, Deep Learning, Computer Vision, Explainable AI (XAI).

VOLUNTEER & LEADERSHIP

Competitive Programming Instructor,

[BUBT CP Community](#)

August 2024 – April 2026

- Mentored 50+ students in C/C++ programming, data structures, and algorithms, focusing on optimization techniques for ICPC-style problem solving.
- Organized coding contests for junior members on virtual judge platforms to improve university-wide contest participation and ranking.